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Report

Mating Regulates Neuromodulator Ensembles at Nerve Termini Innervating the *Drosophila* Reproductive Tract

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Summary

Upon mating, regions of the female reproductive tract mature and alter their function [1–3], for example to facilitate storage of sperm or control the release of eggs [4–6]. The female's nervous system and neuromodulators play important roles in her responses to mating [7–13]. However, it is difficult to reconcile the reproductive tract's many changing but coordinated events with the small set of neuromodulators present [14–18]. We hypothesized that each part of the reproductive tract contains a characteristic combination of neuromodulators that confer unique identities on each region and that postmating changes in these combinations coordinate subsequent actions. We examined the presence, locations, and levels of neuromodulators and related molecules ("signaling molecules") in the reproductive tract of *Drosophila melanogaster* females before and after mating: the biogenic amine octopamine, which regulates ovulation rate in *Drosophila* and locusts [7, 14–20]; serotonin, which regulates muscle contraction in locust oviducts [21]; and the FMRF amide dromyosuppressin, which regulates contraction of *Drosophila* heart muscle [22] and may regulate muscle contractions in the reproductive tract, if it is expressed there. We find that separate aspects of mating (sperm, seminal proteins, and physical effects) independently modulate the release of signaling molecules. Each reproductive tract subregion displays a characteristic combination of signaling molecule release, resulting in a unique functional identity. These patterns, and thus functions, change reproducibly after mating. Thus, one event (mating) promotes new combinations of signaling molecules that endow different parts of the reproductive tract with unique temporal and spatial identities that facilitate many aspects of fertilization.

Results and Discussion

Signaling Molecules Are Distributed in Characteristic Patterns in the Unmated Female Reproductive Tract

To test our hypothesis that combinatorial effects of signaling molecules drive the coordinated changes in the female reproductive tract after mating, we used whole-mount immunocytochemistry to examine the presence, locations, and levels of signaling molecules in the reproductive tracts of unmated females. Octopamine, serotonin, and dromyosuppressin (DMS) immunoreactive nerve termini were found in all regions of the reproductive tract. Although these molecules often colocalized, each had a unique distribution throughout the tract (Figures 1A and 1B). The highest fluorescence intensity for octopamine was in nerve termini of the lateral oviducts (Figure 1B; 3.5-fold higher than in the ovary, $p = 0.02$), for serotonin, in the seminal receptacle (1.6-fold higher than in the ovary, $p = 0.45$), and for DMS, in the lateral oviducts (1.6-fold higher than in the uterus, $p = 0.05$). High fluorescence intensity for a molecule indicates inhibition of its release, or stimulation of its accumulation, in the secretory granules at nerve termini. Conversely, decreased intensity indicates release [8]. Our results show that, for example, the oviducts of unmated females have low net release of octopamine. However, each region of the reproductive tract contains a unique, region-specific combination of signaling molecules that could potentially induce specific functions for that region.

In addition to the three molecules examined here whose levels we could measure directly, other signaling molecules may contribute to the region-specifying ensembles. The diffusible signaling molecule nitric oxide (NO) is important in reproductive tract contractions in mammals [23, 24], and we show that levels of nNOS, the enzyme that synthesizes NO, differ throughout the reproductive tract (see the Figure S1 available online). Another potential contributor is dopamine, which affects ovarian development [25].

We next focused on the common oviduct to compare the spatial distributions of molecules at a finer level. Its linear and simple shape allowed us to precisely align oviducts from different females (Figure 2). We developed a quantitative image analysis method to generate objective comparisons across samples (see the Supplemental Experimental Procedures and Figure S3A). We applied an intensity threshold followed by a Gaussian fit that allows identification of the precise position and intensity of the source of each immunofluorescence signal. We collapsed the 3D (cross-sectional) data into 1D to assess the intensity of each signaling reporter along each anterior/posterior (A/P) position in each oviduct. This method allows meaningful comparisons between the distributions of different signaling molecules.

We divided the oviduct into four equal subregions along its A/P axis (Figure S3A) and measured the fluorescence intensity in each subregion in cross-section (dorsal/ventral axis) (see the Experimental Procedures). This high-spatial-resolution analysis showed that each signaling molecule has a unique distribution and level along the common oviduct (Figure 2). For example, octopamine immunoreactivity was highest in nerve termini innervating the posterior part, lowest at the anterior (Figures 2A and 3, subregions 4 and 1), and intermediate in the areas in the middle (Figure 3, subregions 2 and 3). In contrast, serotonin immunoreactivity intensity was similar at nerve termini innervating the anterior and posterior parts (Figures 2B and 3, subregions 1, 2, and 3). Levels of nNOS also varied along the oviduct (Figures S2A and S3B).

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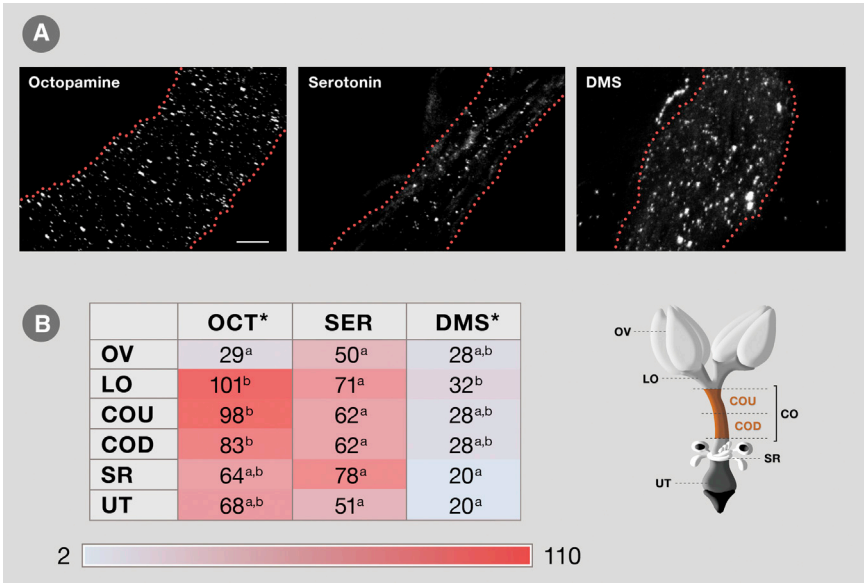


Figure 1. Different Signaling Molecules Innervate the Reproductive Tracts of Unmated Females

(A) Distribution of octopamine, serotonin, and dromyosuppressin immunoreactivity in the unmated female common oviduct. Note the difference in the spatial distribution of the different signaling molecules along the common oviduct (whose outline is shown with red dots). Schematic of the female reproductive system shows the common oviduct (CO) region presented. Scale bar, 50 μm (for all panels).

(B) Heatmap illustrating the relative immunoreactivity of signaling molecules at different regions of the unmated female reproductive tract (ovary [OV], lateral oviducts [LO], upper common oviduct [COU], lower common oviduct [COD], seminal receptacle [SR], and uterus [UT]). Schematic of the female reproductive system shows reproductive tract regions presented. One-way ANOVA, *p < 0.05; Duncan post hoc ranking test. Letters represent grouping.

See also Figure S1.

Mating Mediates Changes in Octopamine, Serotonin and DMS Levels at Reproductive Tract Nerve Termini

To test whether the combination of signaling molecules changes after mating, we compared octopamine, serotonin, and DMS immunoreactivity between unmated females and females at 20, 90, and 180 min after the start of mating (ASM) to wild-type (WT) males. These three time points correspond to the end of copulation, the start of ovulation and peak of sperm storage, and the start of egg deposition, respectively. Our results provide the first dynamic view of mating-induced changes in the reproductive tract's neuromodulatory system.

Mating induced changes in the levels and distribution of signaling molecules in all regions of the female reproductive tract (Figures 4A and S4A and Table S1); these changes were visible by the end of mating. There was a significant decrease in serotonin immunoreactivity in the ovary (2.1-fold, p = 0.019) and seminal receptacle (2.2-fold, p = 0.019) (Figure S4A and Table S1). DMS immunoreactivity decreased in the lateral oviducts (1.23-fold, p = 0.023) and in the upper (COU) and lower (COD) common oviduct (2.3-fold, p < 0.001) and increased in the ovary (1.34-fold, p = 0.041) and uterus (2.75-fold, p = 0.008) (Figure S4A and Table S1). The decrease in serotonin and DMS immunoreactivity soon after mating could indicate their release from vesicles or rapid transport of the neuromodulator into nerve termini [8, 26, 27]. Thus, various reproductive tract tissues are differentially exposed to changing levels or combinations of signaling molecules soon after mating.

At 90 min ASM, the time of peak sperm storage and ovulation onset [7, 28, 29], octopamine immunoreactivity decreased relative to unmated controls in the seminal receptacle (4.3-fold, p = 0.003) and uterus (2.9-fold, p = 0.0041) (Figure 4A and Table S1). Serotonin immunoreactivity decreased in the ovary (1.77-fold, p = 0.018) but increased in the common oviduct (COU and COD: 1.76-fold, p = 0.014) (Figure S4A and Table S1). DMS immunoreactivity decreased in the ovary and in the common oviduct (COU: 2.2-fold, p < 0.001; COD: 4-fold, p = 0.023) and increased in the uterus (2.7-fold, p < 0.001) (Figure S4A and Table S1). These results indicate that mating does not simply trigger a single short-term change, but initiates an orchestrated and progressive change in the level and distribution of signaling molecules in each region of the tract.

By 180 min ASM, we observed additional changes in all regions of the reproductive tract except for the seminal receptacle and uterus. Octopamine immunoreactivity decreased in the lateral oviducts (2-fold, p = 0.027), upper common oviduct (2.2-fold, p = 0.05), seminal receptacle (4.3-fold, p = 0.003), and uterus (2.9-fold, p = 0.045) relative to levels in unmated females (Figure 4A and Table S1). Serotonin immunoreactivity decreased at nerve termini innervating the ovary (2.26-fold, p = 0.004) and lateral oviducts (2.97-fold, p = 0.015) (Figure S4A and Table S1). DMS signals decreased in the ovary (1.9-fold, p < 0.001) and common oviduct (1.3-fold, p = 0.001) and increased in the uterus (2.4-fold, p = 0.002) (Figure S4A and Table S1). These results show that by 180 min ASM, serotonin and DMS were released or stopped accumulating only in the upper reproductive tract. In contrast, in the lower reproductive tract only octopamine continued to be released in the seminal receptacle and uterus.

Mating also changed the distribution of signaling molecules at a finer scale, as seen in subregions of the common oviduct (Figures 2 and 3). Octopamine immunoreactivity increased in nerve termini innervating the anterior common oviduct by 20 min ASM (Figure 2A versus Figure 2D; Figure 3, subregion 1). By 90 min ASM (when the first ovulated egg is detected), octopamine immunoreactivity had decreased in nerve termini innervating the common oviduct and showed no pronounced differences between subregions (Figure 2A versus Figure 2J; Figure 3). Changes in octopamine signal at this time are consistent with the finding that this neuromodulator is essential for ovulation [18, 30] and the timing of onset of increased ovulation after mating [28], as well as with Kapelnikov et al.'s [2] finding that the number of octopaminergic (type II) boutons is stimulated by mating.

As we saw for the whole reproductive tract, other signaling molecules also changed in distribution and level at the fine scale in the common oviduct. At 20 min ASM, DMS immunoreactivity increased greatly in nerve termini innervating the posterior part (Figure 2C versus Figure 2F; Figure 3, subregions 3 and 4). At 90 min, serotonin immunoreactivity changed from uniform across the common oviduct to different in each

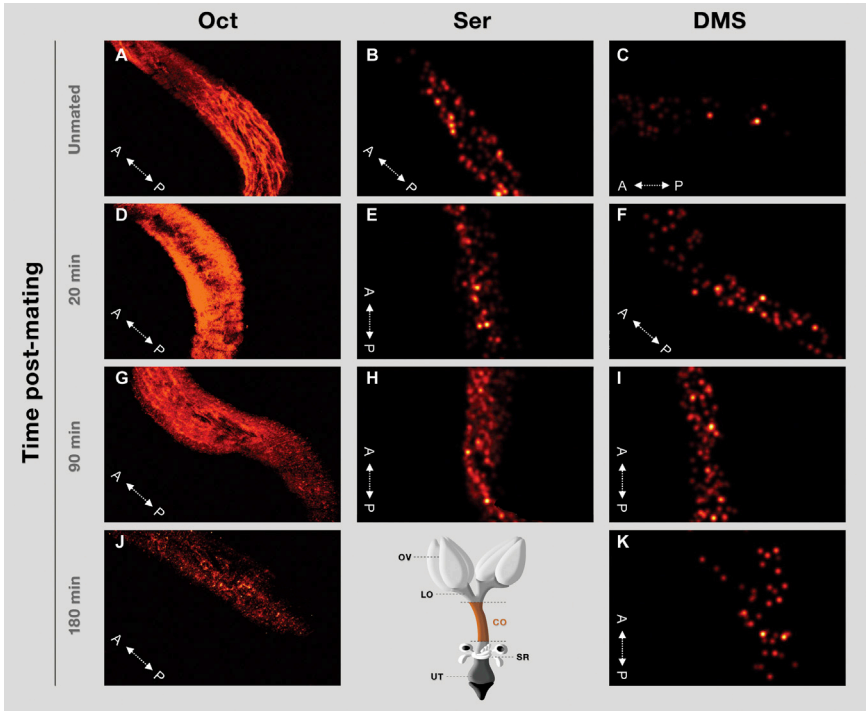


Figure 2. Mating Changes the Spatial Distribution of Signaling Molecules along the Common Oviduct

Representative images of analyzed distribution of octopamine (Oct; A, D, G, and J), serotonin (Ser; B, E, and H), and dromyosuppressin (DMS; C, F, I, and K) along the common oviduct of unmated females (A–C) and females mated to WT males at 20 (D–F), 90 (G–I), and 180 (J and K) min ASM analyzed using quantitative image processing (see the Supplemental Experimental Procedures). The representative image presented for each signaling molecule is an artificial image whose number of spots represents the number of spots above the threshold in the original image. The intensity of each spot is represented on a black-red-yellow (yellow is the highest) color scheme. White arrows indicate the A/P axis (A, anterior; P, posterior). Seven to ten female reproductive tracts were examined for each treatment at each time point. The schematic of the female reproductive system shows the common oviduct (CO) region presented.

subregion (e.g., Figure 2B versus Figure 2H; Figure 3, subregion 2 versus subregion 3). Changes in the level of NO signaling are also most likely induced (see Figures S2A and S3B).

Thus, mating clearly creates new, region- and time-specific ratios of signaling molecules in the reproductive tract, suggesting a way for a small number of molecules to coordinate the activities of different regions of the organ system after mating. Because different signaling molecules often act through different secondary messenger systems, altered ratios of released signaling molecules might cause combinatorial effects on target cells essential to coordinate the functions of the different reproductive tract organs and regions over time. For example, tyramine and octopamine have opposite effects on the locomotory behavior of *Drosophila melanogaster* larvae, and their balance is required for normal locomotive behaviors [31]. Similarly, oviduct muscle contractions are inhibited by octopamine but enhanced by proctolin [32]. The mechanisms by which these dynamic changes could be accomplished in the reproductive tract after mating will be important areas of future investigation.

Seminal Proteins and Sperm Differentially Mediate Postmating Changes in Signaling Molecule Distributions in the Female Reproductive Tract

Mating introduces several triggers into females: sperm, seminal proteins, and other molecules, as well as physical effects [33]. Each of these male-supplied components can affect female reproductive tract responses [5, 8, 34]. We examined how octopamine, serotonin, and DMS distributions were influenced when females were mated to males lacking specific seminal fluid components (see Figures 4 and S4B and Tables S2 and S3).

At 20 min ASM, ovarian octopamine immunoreactivity did not differ between females mated to spermless males and those mated to control (WT) males—so sperm do not regulate ovarian octopamine at this time. However, there was significantly more ovarian octopamine immunoreactivity in females mated to DTA-E males (who lack both sperm and Acp seminal proteins from the males' accessory glands), compared to the levels in mates of WT males. Mates of DTA-E males also showed significantly less octopamine immunoreactivity in nerve termini innervating the uterus compared to the levels in mates of WT or spermless males ($p < 0.05$; Figure 4B; Tables S2 and S3). Thus, Acps stimulate octopamine release at nerve termini innervating the ovary by 20 min ASM and cause octopamine accumulation at uterine nerve termini (potentially by inhibiting release) at this time; the octopamine receptor OAMB is expressed in positions consistent with this effect [16, 17, 30]. Recent findings that the Acp ovulin stimulates ovulation by inducing octopaminergic signaling [7] and that OA/*dsx*$^{+}$ neurons from the abdominal ganglion mediate multiple postmating responses [35] provide functional correlates of the changed distributions that we report here; our present results suggest, for example, that at least some of ovulin's effects are on octopaminergic neurons of the ovary. At this same time point, ovarian serotonin levels did not differ between females mated to either spermless or DTA-E males and those mated to WT males (Figure S4B and Table S2). Thus, neither Acps nor sperm cause the change in serotonin levels at this time; the act of mating and/or seminal fluid components other than Acps or sperm must mediate these changes.

Sperm are needed for the changes seen in octopamine levels at 90 min ASM in nerve termini innervating the lateral oviducts: mates of DTA-E and spermless males showed significant differences in octopamine immunoreactivity compared to mates of WT males ($p < 0.05$; Figure 4B and Table S2), but the differences in octopamine levels between DTA-E and spermless mates were not significantly different. At this time point, serotonin levels at the ovary and common oviduct were nearly identical in mates of spermless, DTA-E, and control males (Figure S4B and Table S2). Thus, mating components other than Acps and sperm regulate serotonin levels and distribution in the ovary and common oviduct at this time (Figure S4B and Table S2).

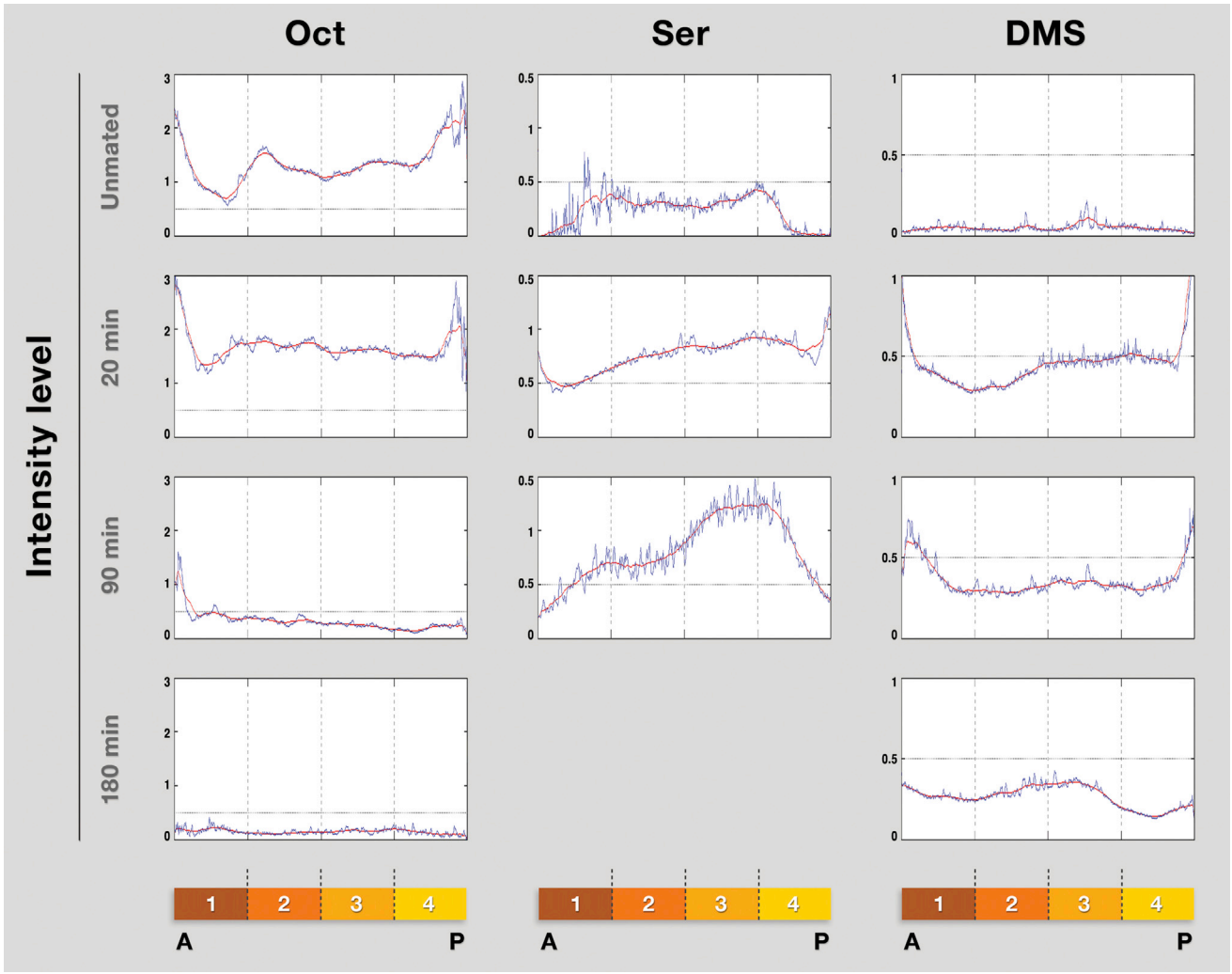


Figure 3. Signaling Molecules Show Distinct Spatial Patterns in Subregions of the Common Oviduct, and These Patterns Are Altered after Mating

Quantitative image processing analysis of octopamine (Oct), serotonin (Ser), and dromyosuppressin (DMS) in four different parts of the common oviduct (brown, 1; dark orange, 2; light orange, 3; and yellow, 4; see the spatial analysis section in the Supplemental Experimental Procedures and Figure S3A) along the A/P axis (A, anterior; P, posterior) of unmated and mated common oviducts at 20, 90, and 180 min ASM. Seven to ten female reproductive tracts were examined for each treatment at each time point. The scale shows the color code for each common oviduct (CO) region presented. See also Figure S3.

The same logic indicates that by 180 min ASM, changes in octopamine levels in the common oviduct and uterus are no longer regulated by Acps or sperm. Rather, mating and/or seminal fluid components other than Acps and sperm are essential for the changes in octopamine levels (Figure 4B and Table S2). Changes in serotonin levels in the ovary and lateral oviducts are also regulated by mating and/or seminal fluid components other than Acps and sperm (Figure S4B and Table S2).

Conclusions

A complex organ system like a reproductive tract must coordinate its various parts to function efficiently. Ovulation and egg movement through the tract must be coordinated with the release of sperm from storage, and the time during which the egg resides in the uterus also needs to be regulated. Neuromodulators regulate these postmating responses [7, 35]. Here, we show that each region of the female reproductive tract in *Drosophila* has a characteristic combination of signaling molecules that impact neural activity, giving each region a unique "neuromodulator combinatorial characteristic." The process of mating introduces several components into the female reproductive tract (sperm, seminal proteins, and potentially physical effects) that differentially and independently affect the distribution patterns and released levels of particular signaling molecules. As a result, the combination and levels of signaling molecules in each region of the reproductive tract changes with time, including at a fine scale, after mating, permitting the coordination of sequential events in this organ system. The male components could alter the combination of neuromodulators by acting directly on cells in the female reproductive tract or indirectly by affecting the female's central nervous system [9, 11–13, 36]. Furthermore, a given signaling molecule in the female reproductive tract can be regulated by different mating components spatially and temporally. For example, octopamine immunoreactivity in the uterus is controlled by Acps immediately after mating, but by 180 min ASM other components control octopamine

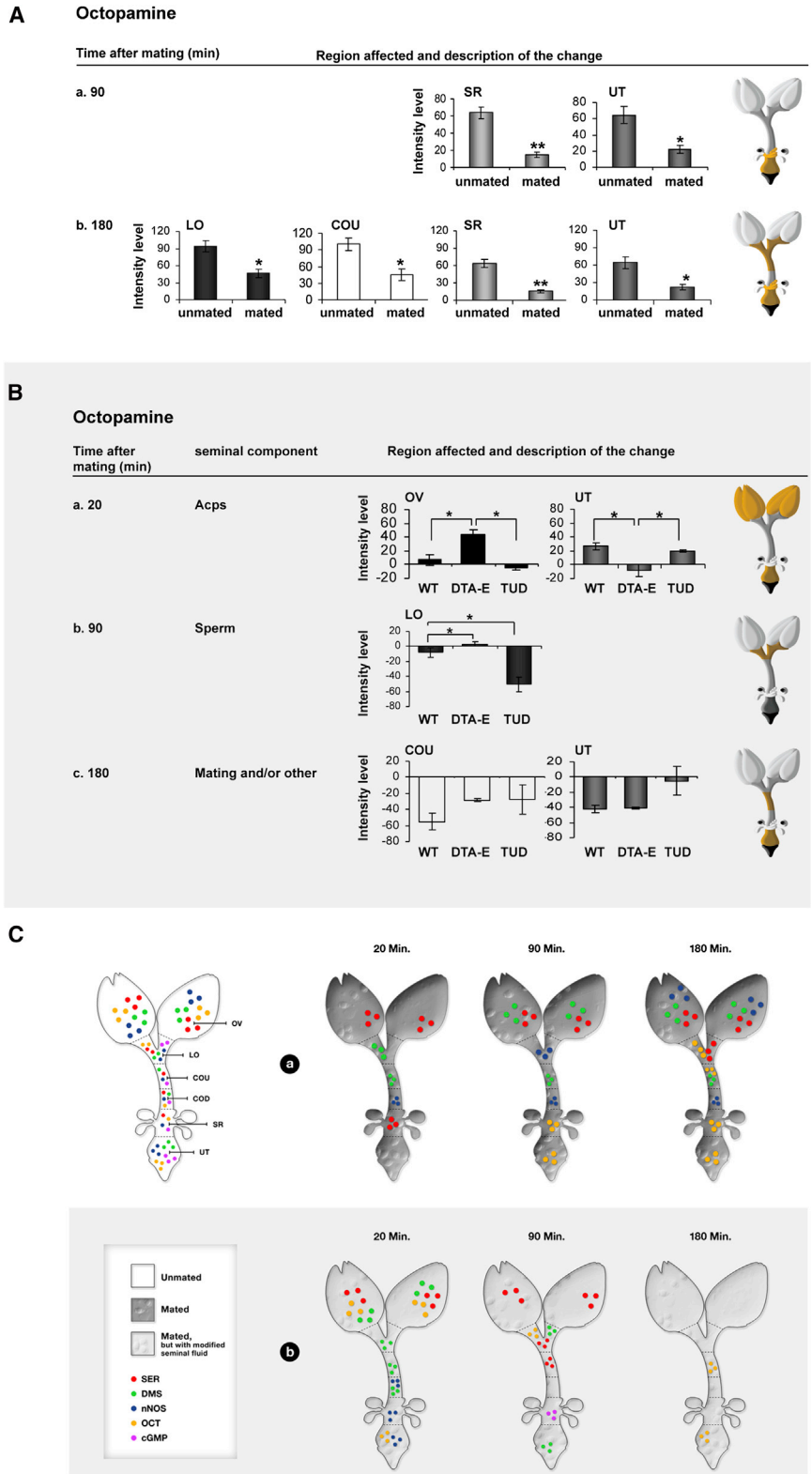


Figure 4. Shortly after the Start of Mating, Octopamine Fluorescence Intensity Levels Significantly Decrease at Nerve Termini Innervating the Lower Reproductive Tract

(A) Graphs are shown only for times postmating and regions in which significant changes were observed. Shown are (Aa) reproductive tract regions (seminal receptacle [SR] and uterus [UT]) with differences in octopamine fluorescence intensity level between unmated females and mates of WT males at 90 min ASM and (Ab) lateral oviducts (LO), lower common oviduct (COD), SR, and UT at 180 min ASM. Reproductive tract fluorescence was quantitated in Image J (see the Supplemental Experimental Procedures). Plotted are means ± SEs of unmated and females mated to WT males. Two independent-sample t tests, *$p < 0.05$, **$p < 0.01$. The reproductive tract schematics are colored to show where the changes occur. Thirty-five to 40 female reproductive tracts were examined for each treatment at each time point.

(B) Specific aspects of mating mediate octopamine vesicle release at different regions of the female reproductive tract and at different time post mating. (Ba) Acps induce immediate (20 min) postmating changes in octopamine immunoreactivity at nerve termini innervating the ovary (OV) and the uterus (UT). (Bb) Acps and sperm induce postmating changes in octopamine immunoreactivity at nerve termini innervating the lateral oviducts (LO) at 90 min ASM. (Bc) Later (180 min), changes in octopamine immunoreactivity at nerve termini innervating the lower common oviduct (COD) and the uterus are induced by the physical act of mating and/or other seminal fluid components other than Acps and sperm. Values of octopamine fluorescence intensity level are plotted as means ± SEs of mates of WT, DTA-E, and spermless (TUD) males. The plotted means are minus unmated female mean values. *$p < 0.05$ indicates, for example, that octopamine fluorescence intensity level in the specific region examined of females mated to TUD males is significantly different from that of females mated to DTA-E males (see Table S3). Schematics of the female reproductive system are colored to show where the changes occur. We only show cases where statistically significant differences were observed between mates of WT and mates of TUD and/or DTA-E males. Thirty five to 40 female reproductive tracts were examined for each treatment at each time point.

(C) A model that illustrates signaling molecules networks in different regions and subregions along the female reproductive tract pre- and postmating. The schematic of the unmated female reproductive tract shows the different regions and spatial localization of signaling molecules (or pathway members). These molecules are represented by colored dots (red, serotonin [SER]; green, dromyosuppressin [DMS]; orange, octopamine [OCT]; blue, neuronal nitric oxide synthase [nNOS]; pink, cyclic guanosine monophosphate [cGMP]); colors are as in (A) and (B) and Figures S4A and S4B. (Ca) shows schematics of female reproductive tracts after mating to males with normal (WT) seminal fluid (dark gray background). Mating induces a change in signaling molecules' levels along the reproductive tract. The dots represent the signaling molecules whose level changes in each given region at 20, 90, and 180 min ASM. (Cb) shows schematics of female reproductive tracts after mating to males with modified seminal fluid (light gray background). Acps, sperm, and mating and/or other seminal components mediate changes in signaling molecules in the female reproductive tract postmating. The dots represent signaling molecules whose level is regulated by Acps, sperm, Acps and sperm, or other mating components in the different regions at 20, 90, and 180 min ASM.

See also Figure S4 and Tables S1–S3.

levels (Figures 4B and 4C). Finally, effects of mating components on signaling molecule levels are seen most strongly at 90 min ASM. This suggests that after this time, the effects of mating components have begun to decrease (Figure 4C) because the spatial arrangement and/or functional identity of the reproductive tract subregions has already been established or because female components are beginning to exert greater control over these arrangements or functional identities. Moreover, since the presence or level of one neuromodulator can modify the response to another [31, 32], simultaneous analysis of the dynamics of multiple signaling molecules and their interaction with individual male-supplied molecules, such as seminal fluid proteins, is required to fully understand the function and regulation of a complex organ system like the reproductive tract.

Experimental Procedures

Fly handling, reproductive tract preparation, immunocytochemistry, and confocal microscopy and its quantitation with Image J were all performed by standard methods (see the Supplemental Experimental Procedures for details). Methods for evaluation and comparison of spatial fluorescence intensity, including quantitative image processing and image analysis, are described in detail in the Supplemental Experimental Procedures.

Supplemental Information

Supplemental Information includes Supplemental Results and Discussion, Supplemental Experimental Procedures, four figures, and three tables and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.042.

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